SAKSHAM PANDEY

Full Stack Software Engineer | Distributed Systems & Web Platforms

Patna, India (Open to Relocation to Germany / Remote)
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Work Authorization: Indian National — Eligible for Germany IT Specialist Visa / EU Blue Card

PROFESSIONAL SUMMARY

Performance-driven Full Stack Engineer with 3+ years of production experience designing resilient distributed systems, real-time communication protocols (WebRTC, WebSocket), and high-throughput web backends. Skilled in React, TypeScript, Python (FastAPI), Node.js, and relational database internals (PostgreSQL). Demonstrated track record reducing query latency by 76%, sustaining 99.8% production uptime, and shipping containerized microservices via automated CI/CD pipelines.

TECHNICAL SKILLS

Core Languages: TypeScript, JavaScript (ES6+), Python, SQL, Go (Basics), HTML5, CSS3
Frontend Engineering: React.js, Next.js, Redux Toolkit, Context API, Tailwind CSS, WebRTC, WebSockets
Backend Architecture: FastAPI, Node.js, Express.js, Flask, RESTful APIs, Microservices, Redis Caching, OAuth 2.0, JWT
Databases & Storage: PostgreSQL, MySQL, Redis, Relational Schema Design, Query Optimization & Indexing
DevOps & Infrastructure: Docker, Git, GitHub Actions, CI/CD Pipelines, Linux/Bash, Vercel, Render, Postman, Agile/Scrum

PROFESSIONAL EXPERIENCE

Software Engineer (Contract & Independent Engineering)

Remote | 01/2023 – Present

- Engineered and deployed 6 commercial full-stack web platforms using React, TypeScript, FastAPI, Node.js, and PostgreSQL, ensuring **99.8% system availability** across production client environments.
- Implemented centralized OAuth 2.0 / JWT authentication and granular Role-Based Access Control (RBAC), remediating OWASP top 10 vulnerabilities (XSS, CSRF) across administrative dashboards.
- Refactored synchronous REST endpoints and introduced composite B-tree indexing in PostgreSQL and MySQL, driving p95 query response times down from **280 ms to 65 ms (76% latency reduction)**.
- Containerized distributed microservices with Docker and built automated testing/deployment pipelines via GitHub Actions, accelerating release cycles by **40%**.

TECHNICAL PROJECTS

Viora Meet – Low-Latency Offline-Aware Video Engine

React • FastAPI • WebRTC • WebSocket | GitHub

- Designed a WebRTC peer-to-peer streaming engine using an asynchronous FastAPI WebSocket signaling layer, maintaining **sub-150 ms** end-to-end media transmission latency.
- Implemented a fault-tolerant local-area network (LAN) fallback mechanism to preserve local mesh audio/video feeds during WAN outages.
- Built dynamic screen sharing, HTML5 Canvas video rendering, and client-side reconnect state machines handling network jitter and packet loss.

ZipHire – Automated ATS Resume Parsing Engine

React • Python (FastAPI) • PostgreSQL • NLP | GitHub

- Developed an automated document parsing engine processing complex multi-page PDF/DOCX layouts through Abstract Syntax Tree (AST) lexical tokenization and NLP extraction pipelines.
- Programmed vectorized semantic search logic matching unstructured candidate profiles against technical job requisitions with **94% validation accuracy**.
- Built tenant-isolated PostgreSQL database schemas and high-throughput data ingestion endpoints with strict RBAC boundary checks.

NexusChat – Event-Driven Real-Time Messaging Platform

React • FastAPI • Redis • MySQL | GitHub

- Designed a bi-directional event-driven messaging service using persistent WebSockets, handling real-time presence indicators and delivery state updates.
- Built cursor-based database pagination combined with Redis caching, supporting zero-latency infinite scrolling over **100k+ stored conversations** without UI frame drops.

EDUCATION

Indian Institute of Technology Patna

Integrated BS-MS in Artificial Intelligence and Cybersecurity | 08/2025 – Expected 05/2030

Core Coursework: Distributed Systems, Data Structures & Algorithms, Network Security, Cryptography, Database Systems, Low-Latency Networking

CERTIFICATIONS & LANGUAGES

- **Certifications:** HackerRank Gold Badge (Python & Problem Solving), Advanced REST APIs & Microservices Architecture
- **Languages:** English (C1 / Full Professional), Hindi (Native), German (A1 – Active Learner), Russian (A2 – Elementary)